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Blender 3D Basics: Student Worksheet

Part 1: Key Terms (Defined in Simple English)

- **3D Viewport:** The primary workspace window where you view, create, and manipulate your 3D models.
- **Mesh:** A 3D object constructed out of connected points, lines, and surfaces (e.g., cubes, cylinders, characters).
- **Vertex (plural: Vertices):** A single point or corner in 3D space.
- **Edge:** A straight line connecting two vertices together.
- **Face:** The flat surface enclosed by edges (e.g., one side of a cube).
- **Object Mode:** The mode where your mesh acts as a single solid unit; used for overall positioning, moving, and scaling.
- **Edit Mode:** The mode where you modify the inner structure of a shape by moving individual vertices, edges, or faces.
- **Extrude:** Pulling out a face, edge, or vertex to generate brand-new, connected 3D geometry.
- **Bevel:** Rounding off or angling sharp edges to make them soft or realistic.
- **Loop Cut:** Slicing a ring of connected edges all the way around a model to provide extra geometry for editing.
- **Axis Locking (X, Y, Z):** Locking movement, rotation, or scaling strictly to one axis (Red = X, Green = Y, Blue = Z).
- **Shade Smooth / Auto Smooth:** A lighting setting that makes angular models look rounded and smooth without adding polygon lag.

Part 2: Essential Blender Shortcuts & Hotkeys

Shortcut / Key	Action	Tips & Context
Middle Mouse Scroll	Zoom In / Out	Zooms viewport toward your cursor.
Middle Mouse Click + Drag	Orbit / Rotate Camera	Spins the camera around the active object.
Shift + Middle Mouse Drag	Pan View	Slides camera view up, down, left, or right.
Alt + Middle Mouse Drag	Snap to Orthographic View	Snaps camera to Front, Side, or Top without numpad.
Numpad 1 / 3 / 7 / 9	Preset Camera Views	1=Front, 3=Right, 7=Top, 9=Bottom view.
Tab	Toggle Mode	Switches between Object Mode and Edit Mode instantly.
G	Grab / Move	Moves the active selection freely.



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S	Scale / Resize	Shrinks or enlarges selection.
R	Rotate	Rotates selection according to current viewport angle.
G / S / R + (X, Y, or Z)	Axis Constraint	Example: "R + Y" rotates purely on the Y-axis.
A	Select All	Selects all vertices, edges, or objects.
L	Select Linked	Selects an entire connected mesh piece under the cursor.
E	Extrude	Pulls new geometry outward from selected face or edge.
E + Z	Extrude on Z-Axis	Pulls new geometry straight up or down vertically.
Ctrl + R	Add Loop Cut	Slices new ring cut. Scroll wheel to add multiple cuts.
Ctrl + B	Bevel Edges	Rounds sharp edges. Scroll wheel to add segments.
Shift + A	Add Menu	Opens menu to add new 3D primitive shapes.
Shift + D	Duplicate	Creates a duplicate clone of the selected mesh.
Ctrl + Z	Undo	Reverts the most recent action.

⚠ Beginner Troubleshooting Tip: If you cannot select an entire single side of your object, check your Selection Mode in the top-left corner! You must be in Edit Mode (Tab) with Face Selection activated (press "3" on your keyboard number row).

Part 3: Step-by-Step Modelling Walkthroughs

Guide A: Switching Selection Modes in Edit Mode

Press Tab to switch from Object Mode into Edit Mode.

Locate the three selection mode icons in the top-left toolbar (or press keys 1, 2, 3 on your top number row):

- Key 1 (Vertex Select): Select individual corner points to push and pull.
- Key 2 (Edge Select): Select lines between points.
- Key 3 (Face Select): Select whole flat polygon surfaces (use this to select whole sides).

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Guide B: Modeling a Low-Poly Pine Tree (Step-by-Step)

Step 1: Create the Trunk

- Delete the starter cube by pressing X or Delete.
- Press Shift + A to open the Add Menu, navigate to Mesh, and select Cylinder.
- In the small settings drawer at the bottom-left of the screen, set Vertices to 8 (this creates clean, low-poly game geometry).
- Press Tab to enter Edit Mode, select Face Select (Key 3), and click the top circle face.
- Press G then Z to pull the top face upward, making a tall tree trunk.
- Press S to scale the top face inward slightly to give the trunk a natural taper.
- Right-click the mesh and select Shade Auto Smooth to clean up the surface lighting.

Step 2: Model the Branch Layers (Leaves)

- Press Shift + A -> Mesh -> Circle. In the bottom-left popup, set Vertices to 12.
- Press Tab to enter Edit Mode, switch to Edge Select (Key 2), and select the circular edge.
- Press E then Z to extrude downwards, creating an open cylinder sleeve.
- Press S to scale the bottom edge outward into a flared cone/skirt shape.
- Press Ctrl + R to place a Loop Cut in the middle of this skirt, click to confirm, and press S to flare the middle.
- Hover your mouse over this leaf tier and press L to select the entire connected piece.

Step 3: Stacking and Finishing the Tree

- Press Shift + D to duplicate the foliage tier and drag it up along the trunk.
- Press S to scale this higher tier slightly smaller, and press R to rotate it slightly for natural variation.
- Duplicate (Shift + D) and scale down (S) 2 more times until you reach the top of the tree.
- Select the finished tree, right-click, and select Shade Auto Smooth.